

WASP-43b

R-0636

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_170415 · created 2026-07-25T17:43:33+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457858.6562 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.09 +/- 0.053
Transit depth (Rp/Rs)^2	0.82 +/- 0.97 [%]
Orbital Inclination (inc)	86.96 +/- 2.96
Airmass coefficient 1 (a1)	0.566 +/- 0.062
Airmass coefficient 2 (a2)	0.39 +/- 0.24
Scatter in the residuals of the lightcurve fit is	11.03 %
Best Comparison Star	#2 - [543, 191]
Optimal Method	PSF photometry
Transit Duration (day)	0.0549 +/- 0.0054

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.09 ± 0.053 vs 0.1594 (deviation 1.19σ)
Duration fit vs published	0.0549 d vs 0.0512 d (deviation 0.61σ)
Mid-time O-C	-3.2 min
Depth SNR	1.5
Residual scatter	11.03 %

Light curve

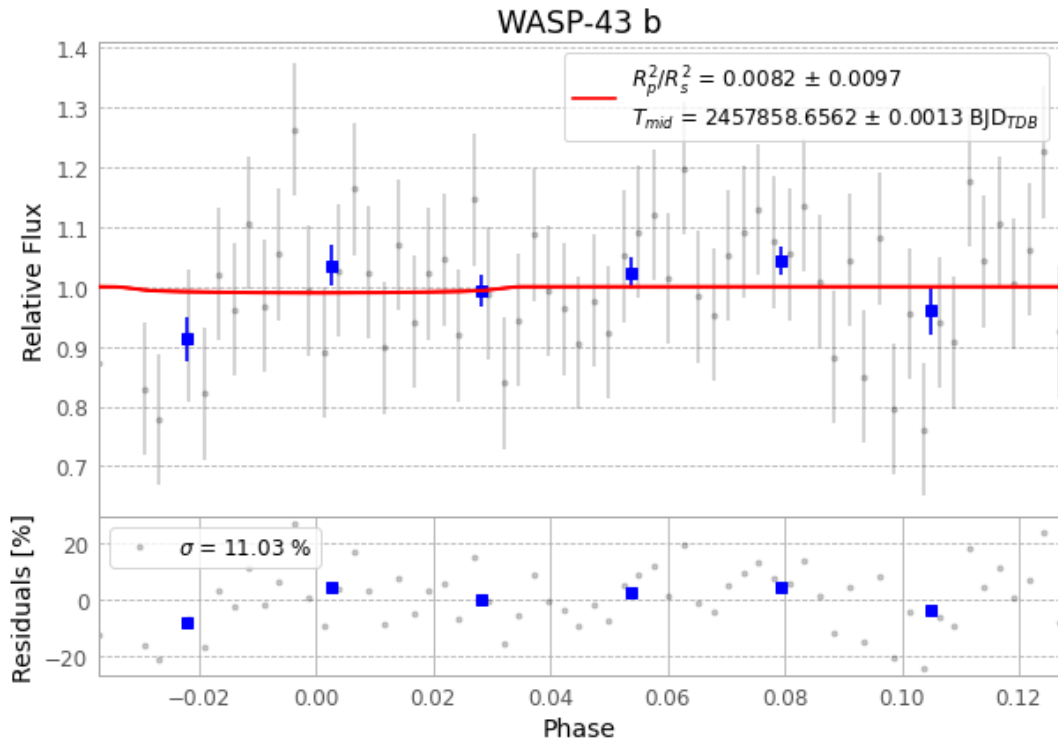


Figure 1 — FinalLightCurve_WASP-43 b_2017-04-14.png (SHA-256 4ba4b5dd8758810b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [336, 167], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 50166b052aeafd70...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

63 FITS frames (41 MB), WASP-43170415025414.FITS → WASP-43170415060612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-170415.log (a3417d405286...), FinalLightCurve_WASP-43 b_2017-04-14.png (4ba4b5dd8758...), FinalLightCurve_WASP-43 b_2017-04-14.csv (5edf2403c195...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 17:43Z.